

1. Administrative & Study Identification

Full Study Title: NIS to Examine the Effectiveness of TDC in Patients With Metastatic Non-squamous NSCLC and High-risk Genetic Alterations **Short Title/Acronym:** NAUTIC **Protocol Number:** D419MR00003 **NCT Number:** NCT06494540 **EudraCT Identifier:** N/A **CTIS Identifier:** N/A **Sponsor Name:** AstraZeneca **Investigational Product:** TDC (tremelimumab and durvalumab in combination with a platinum-based chemotherapy) **Phase of Development:** N/A (Observational / Non-interventional) **Overall Status:** Recruiting **Verification Date:** 01-Mar-2026 **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To describe the real-world effectiveness of TDC with respect to high-risk genetic alterations (mutations in KRAS, STK11, KEAP1, and TP53) as well as the expression of TTF-1 and PD-L1 in routine clinical practice. * **Secondary Objectives:** To deepen the understanding of optimal, biomarker-guided treatment strategies for non-squamous metastatic non-small cell lung cancer (NSQ mNSCLC) in distinct subgroups with a high medical need.

2.2 Background & Rationale While combination immunotherapies such as Tremelimumab and Durvalumab with chemotherapy (TDC) are approved for first-line treatment of mNSCLC, real-world evidence is needed to better understand their effectiveness in specific high-risk mutational subgroups. This study collects real-life data in Germany to evaluate how biomarkers like KRAS, STK11, KEAP1, and TP53 impact patient outcomes outside of controlled clinical trial environments.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Observational / Non-interventional Study (NIS) * **Allocation:** N/A * **Intervention Model:** N/A * **Endpoint Classification:** N/A * **Primary Purpose:** N/A * **Control Type:** Single-arm * **Blinding / Masking:** N/A / Open-label (Routine Clinical Practice) * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Estimated Enrollment:** 600 patients * **Number of Sites:** Multicenter in Germany * **Study Start Date:** 28-Jun-2024 * **Estimated Primary Completion Date:** 30-Jun-2028 * **Estimated Study Completion Date:** 30-Jun-2028

4. Subject Selection (Eligibility Criteria)

Demographics: * **Age:** 18 Years to 120 Years * **Sex:** All * **Healthy Volunteers:** No

4.1 Inclusion Criteria 1. Aged 18 to 120 years. 2. Histologically or cytologically confirmed diagnosis of NSQ mNSCLC (incl. large cell neuroendocrine carcinoma [LCNEC] if considered NSCLC-like by the treating physician). 3. Decision to start first-line (1L) treatment with TDC according to the current Summary of Product Characteristics (SmPCs). 4. Molecular Next Generation Sequencing (NGS) panel as per institutional standard has been initiated (including KRAS, STK11, KEAP1, and TP53) alongside TTF-1 and PD-L1 expression analysis. 5. No sensitizing epidermal growth factor receptor (EGFR) mutations or anaplastic lymphoma kinase (ALK) alterations.

4.2 Exclusion Criteria 1. Current participation in interventional clinical trials. 2. Contraindications according to current SmPCs. 3. Any active tumor other than metastatic NSCLC.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** TDC (Tremelimumab + Durvalumab + Platinum-based chemotherapy) administered according to the marketing authorization and the treating physician's discretion in routine clinical practice. * **Route of Administration:** Intravenous (standard for TDC). * **Duration of Treatment:** As per institutional standards, patient tolerance, and current SmPCs.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard concomitant medications for mNSCLC symptom management as per routine care. * **Prohibited:** Concurrent participation in interventional clinical trials.

6. Study Timeline & Visit Schedule

(Note: As an observational NIS, the study timeline aligns strictly with routine clinical practice rather than a mandated trial schedule) * **Screening / Baseline:** Informed consent, collection of NGS/Biomarker expression profiles, and initiation of first 1L TDC dose. * **Interim / Routine Care:** Standard oncology follow-ups to collect real-world safety and effectiveness data. * **End of Study:** Up to the estimated primary completion date (June 2028) or patient death.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Overall Survival (OS), defined as the time from the date of the first documented dose of TDC until death due to any cause. * **Measurement Method:** Clinical record review using the Kaplan-Meier technique, categorized by specific genetic subgroups (KRAS, STK11, KEAP1, TP53, TTF-1, PD-L1).

7.2 Secondary & Exploratory Endpoints * Deeper understanding of optimal, biomarker-guided treatment outcomes and clinical responses across the distinct high-risk NSQ mNSCLC subgroups.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Collected from real-world routine clinical practice observations.
 - **Serious Adverse Events (SAEs):** Handled according to standard pharmacovigilance and local German regulatory requirements for marketed products.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Patients meeting all eligibility criteria who receive at least one documented dose of TDC.
 - **Handling of Missing Data:** Any patient not known to have died at the time of analysis will be censored based on the last recorded date on which the patient was known to be alive.
 - **Statistical Methods:** Median OS calculated via Kaplan-Meier technique.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** Conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local guidelines for Non-Interventional Studies (NIS).
 - **Data Collection:** Prospective, real-world data collection from institutional standard-of-care patient files.
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11. Study Identifiers & Governance

- **Primary ID:** D419MR00003
- **Registry Number:** NCT06494540
- **Sponsor/Lead Organization:** AstraZeneca
- **Study Title:** NAUTIC
- **Official Regulatory Title:** NIS to Examine the Effectiveness of TDC in Patients With Metastatic Non-squamous NSCLC and High-risk Genetic Alterations

12. Clinical Framework (Oncology Specific)

- **Medical Condition:** Non-squamous metastatic non-Small-Cell Lung Carcinoma (NSQ mNSCLC)
 - **Diagnosis Threshold:** Histologically/cytologically confirmed, EGFR/ALK wild-type
 - **Medical Methodology:** First-line systemic combination immunotherapy + chemotherapy
 - **Optimization Target:** Biomarker-guided treatment efficacy (focusing on KRAS, STK11, KEAP1, TP53)
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13. Study Procedures & Methodology

- **Management Pathway:** Routine real-world clinical oncology pathway in Germany
 - **Molecular Protocol:** Next-generation sequencing (NGS) and IHC for PD-L1/TTF-1 implemented as institutional standards prior to inclusion
 - **Quality Audit Mechanism:** Standard observational data monitoring and source data verification as dictated by AstraZeneca NIS standards
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to ~4 years (June 2024 – June 2028)
 - **Observation Cadence:** Correlated with standard clinical visits for chemotherapy/immunotherapy cycles
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Insert Name / e.g., PD Dr. Heiko Golpon]	____	[DD-MMM-YYYY]	Clinical Operations Lead	[Insert AstraZeneca Rep Name]	____	[DD-MMM-YYYY]
Lead Statistician	[Insert Statistician Name]	____	[DD-MMM-YYYY]				
